

Safety Evaluation Framework for Autonomous Vehicles:

A Data-Driven Analysis of the Waymo Open Perception Dataset v2

By:

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Key findings at a glance

Vehicle-Vehicle (DD) near-miss rate:
14.7 %

How often two moving vehicles come within 3 seconds of collision. 101 of 1,000 driving segments are high-risk.

Denser traffic means more near-miss events

Above-median traffic density adds 17.4pp to the chance a moment contains a near-miss. The rise comes from more vehicle pairs being present, not from each pair being more dangerous. Individual encounters are no riskier in dense traffic.

Vehicle-Obstacle (DS) near-miss rate:
32.3 %

How often a moving vehicle closes within 3 seconds of stopped traffic. 297 of 1,000 segments are high-risk.

Worst-case gaps fall below safe braking distance

In the most dangerous 1 in 10 segments, the closest vehicle approach leaves only 0.141 seconds to collision [0.126, 0.158], too little for emergency braking to prevent contact.

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1 Executive Summary

This report builds and applies a safety evaluation framework to Waymo urban driving data. The central question is: *how do we measure near-miss risk in a way that is physically meaningful, separates different risk types, and can support deployment and policy decisions?* To this end, we apply a set of metrics that distinguish risk modes, characterize rare but serious event behavior through extreme value theory, and connect to a causal understanding of what drives collision risk in the first place.

Four operational findings.

1. **The dynamic-dynamic (DD) near-miss rate is 14.7 %.** Once both-stationary pairs (annotation artefacts) are excluded and analysis is restricted to pairs where both vehicles are moving, that we refer to as dynamic-dynamic, approximately one in seven frames contains a vehicle-to-vehicle conflict ($TTC < 3s$). 101 of 1,000 segments exceed a 50 % near-miss rate, making these candidates for targeted operational review.
2. **High traffic density causally increases DD near-miss risk by 17.4 percentage points.** This is a causal estimate, not an association: after propensity score matching (PSM) on speed and acceleration, crossing the median density threshold raises the binary DD near-miss probability by +17.4 pp (95 % CI: [+14.8, +20.0]; cluster-robust over 997 segments; matched OR = 4.51; E-value ≈ 3.7 , corrected for common outcome). IPW confirms: +20.3 pp (OR = 5.37), within 3 pp of PSM. As a secondary continuous outcome, the ATE on $\min(TTC^{DD}, 60)s$ is $-16.6s$ [$-18.5, -14.7$]; this shifts to $-7.6s$ when the cap is lowered to 30s, confirming the binary near-miss probability is the more interpretable primary endpoint. Naive standard errors from pair-level bootstrap are $8.9\times$ too narrow (design effect 78.67), because consecutive 10 Hz frames within a segment have a median lag-1 autocorrelation of +0.82. Mediation on the continuous TTC outcome attributes 78 % of the effect to a direct pathway (more pairs entering collision geometry) and 22 % to reduced physical spacing; however, §4 shows this “direct” pathway is largely the order-statistic mechanism (more pairs raise the any-pair near-miss probability mechanically). The primary operational lever is reducing the *number of near-miss opportunities* — through density limits or wider following distance — rather than changing per-pair TTC behaviour in dense scenes.
3. **The safety tail is heavy and below braking distance.** Block-maxima GPD analysis on inverse TTC ($1/TTC$) gives a 1-in-10 segment minimum TTC of 0.141 s [0.126, 0.158]. At this TTC, even emergency braking at 1 g cannot prevent a collision in most approach

geometries. The model is validated against empirical counts: 106 segments fall below the 1-in-10 return level vs. 100 expected under the fitted GPD ($\hat{\xi} = 0.454$, $\hat{\sigma} = 2.633$). This characterizes the distribution of worst-case events across the fleet, independent of any individual incident.

4. **Anomalous frames are driven by sudden heading changes, not speed.** Isolation Forest identifies 2% of frames as anomalous. Heading rate is the strongest predictor of anomaly score, ahead of speed and density.

2 Dataset and Problem Setup

2.1 Data

The Waymo Open Perception Dataset v2 provides 3D LiDAR bounding boxes and per-agent velocity vectors from autonomous vehicle operations in San Francisco, Phoenix, and Mountain View. We use the `lidar_box` component, which gives position (x, y) , velocity (v_x, v_y) , and bounding-box dimensions (length, width) for every tracked agent in each frame.

The corpus covers 1,000 driving segments (798 training, 202 validation), totalling 197,807 frames at 10 Hz (≈ 20 s per segment) and 11,964,433 agent trajectory observations. Raw data is 775 MB in Parquet format; pre-computed metrics add 14 MB. All queries are issued via DuckDB directly on the Parquet files to avoid loading full datasets into memory.

2.2 The near-miss measurement problem

The standard time-to-collision metric is:

$$\text{TTC}_{ij} = \begin{cases} -\frac{\Delta \mathbf{r} \cdot \Delta \mathbf{v}}{|\Delta \mathbf{v}|^2} & \text{if } \Delta \mathbf{r} \cdot \Delta \mathbf{v} < 0 \text{ (converging)} \\ +\infty & \text{otherwise} \end{cases} \quad (1)$$

where $\Delta \mathbf{r} = \mathbf{r}_i - \mathbf{r}_j$ and $\Delta \mathbf{v} = \mathbf{v}_i - \mathbf{v}_j$.

Applied naively across all agent pairs in the Waymo data, this gives a near-miss rate (TTC < 3 s) of **94.9%**. That number is not interpretable as a vehicle collision risk: pedestrians and cyclists at crosswalks frequently satisfy the convergence condition relative to nearby vehicles, producing geometrically valid but operationally irrelevant values. More subtly, even after restricting to vehicles, two problems remain. The point-mass formula counts pairs of vehicles that will never physically intersect. For example, vehicles in adjacent lanes because

it ignores lateral clearance. And it treats a stopped vehicle next to a moving one the same as two converging vehicles, conflating two qualitatively different risk scenarios.

Both problems are addressed in the next section.

3 Safety Metric Design

3.1 2-D bounding-circle TTC

Rather than treating vehicles as point masses, we enclose each vehicle in a circle of radius $r_k = w_k/2$ (where w_k is the LiDAR-measured bounding-box width; median $\bar{w} = 2.08$ m in this corpus). Two vehicles i and j first come into contact at time t^* satisfying

$$|\Delta \mathbf{r} + \Delta \mathbf{v} t|^2 = R_{ij}^2, \quad R_{ij} = r_i + r_j \quad (2)$$

Expanding gives a quadratic $at^2 + bt + c = 0$ with $a = |\Delta \mathbf{v}|^2$, $b = 2(\Delta \mathbf{r} \cdot \Delta \mathbf{v})$, $c = |\Delta \mathbf{r}|^2 - R_{ij}^2$. The discriminant $\Delta = b^2 - 4ac$ is key: if $\Delta < 0$, the projected circles never touch and $\text{TTC}_{ij}^{2D} = +\infty$ regardless of closing speed. This one condition eliminates vehicles in opposing lanes (lateral clearance $> R_{ij}$) and any pair whose velocity vectors are not on a colliding heading. The TTC is then the smaller positive root $t^* = (-b - \sqrt{\Delta})/(2a)$. For nearly-stationary pairs ($|\Delta \mathbf{v}| < 0.3$ m/s) we use the linearised form $t^* = -c/b$ to avoid numerical cancellation. A 50 m range filter is applied before evaluating any pair.

3.2 Dynamic–Dynamic and Dynamic–Static pair classification

After resolving the geometry, a further decision is needed: which pairs to include. Two stopped vehicles cannot collide; their bounding circles may overlap due to LiDAR annotation jitter, not physical risk. So we exclude both-stationary pairs (both vehicles with speed ≤ 0.5 m/s) entirely.

The remaining pairs fall into two distinct risk modes:

- **Dynamic–Dynamic (DD):** both vehicles moving (speed > 0.5 m/s). This is the classical vehicle-conflict scenario: head-on, merging, overtaking. The relevant question is whether two vehicles on independent trajectories are converging.
- **Dynamic–Static (DS):** one vehicle moving, one stopped. This measures obstacle proximity: a moving vehicle approaching parked or queued traffic. It is a distinct operational problem: the stopped vehicle poses no independent threat, but dense urban parking creates a persistent spatial constraint on moving vehicles.

Mixing DD and DS into a single metric obscures both. A high combined near-miss rate could mean lots of vehicle-to-vehicle conflicts, or it could mean the route runs through a neighbourhood with dense street parking. They require different interventions.

3.3 Results

Table 1: 2-D bounding-circle TTC by pair type across 197,807 frames and 1,000 segments. Both-stationary pairs excluded.

	Vehicle conflict (DD)	Obstacle proximity (DS)
Near-miss rate ($TTC < 3\text{ s}$)	14.7 %	32.3 %
Median segment minimum TTC	4.39 s	1.70 s
Mean near-miss pairs per frame	0.29	0.89
Segments with nm rate $> 50\%$	101	297

The reduction from a naive 94.9 % to a DD rate of 14.7 % reflects how much of the apparent near-miss signal in urban AV data is attributable to annotation artefacts and mixed risk types rather than genuine vehicle conflicts. The DS rate of 32.3 % is separately meaningful: it characterises how frequently moving vehicles are operating in close proximity to stopped traffic, which affects route planning and speed profile decisions independently of vehicle-to-vehicle conflict risk.

Figure 1 shows the DD and DS TTC distributions across segments and frames.

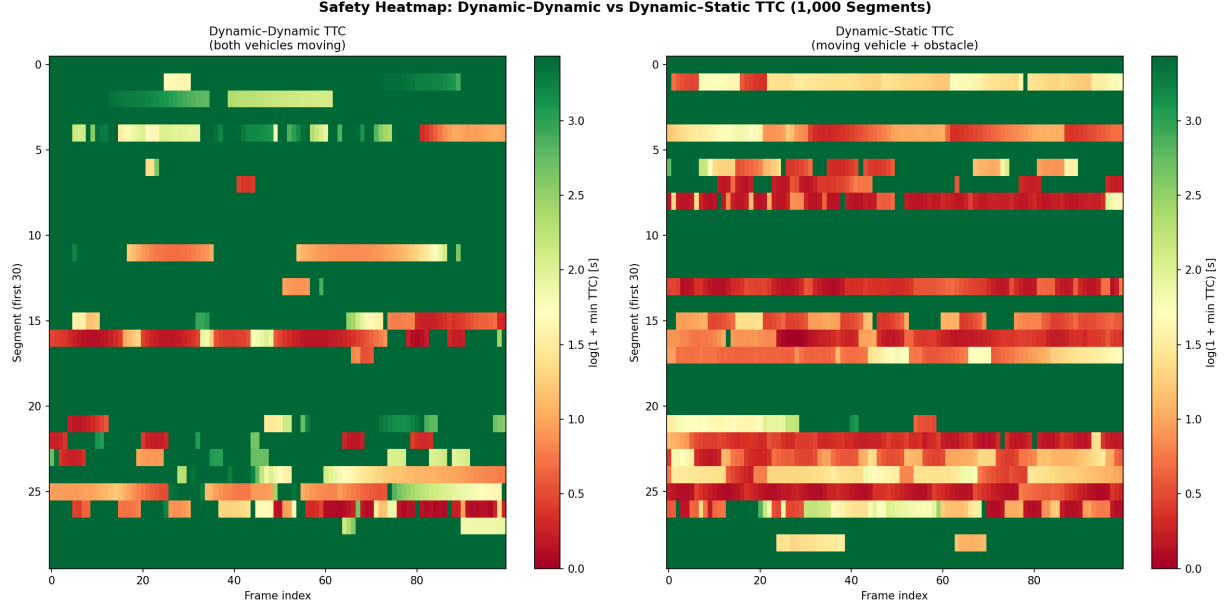


Figure 1: DD (left) and DS (right) minimum TTC across 30 representative segments. Colour encodes $\log(1 + \text{TTC})$; warmer colours indicate lower TTC (higher risk). DS patterns are warmer, reflecting the prevalence of stopped vehicles in urban environments.

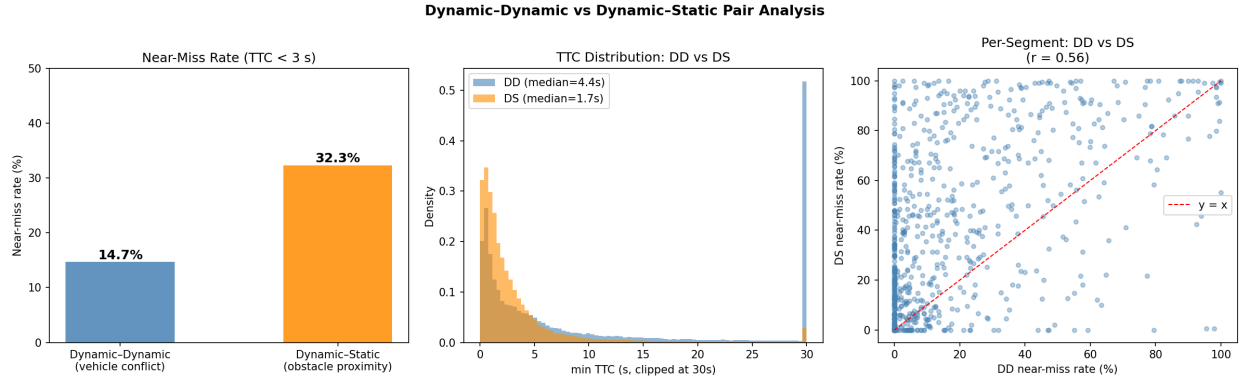


Figure 2: Left: DD vs DS near-miss rates across the corpus. Centre: TTC distributions for DD (blue) and DS (orange); the DS distribution is shifted substantially left. Right: per-segment DD vs DS near-miss rate scatter ($r = 0.31$), showing that the two metrics are partially but not fully correlated — segments with high DD risk do not always coincide with high DS risk.

Figure 3 illustrates how the DD metric behaves within a single representative segment.

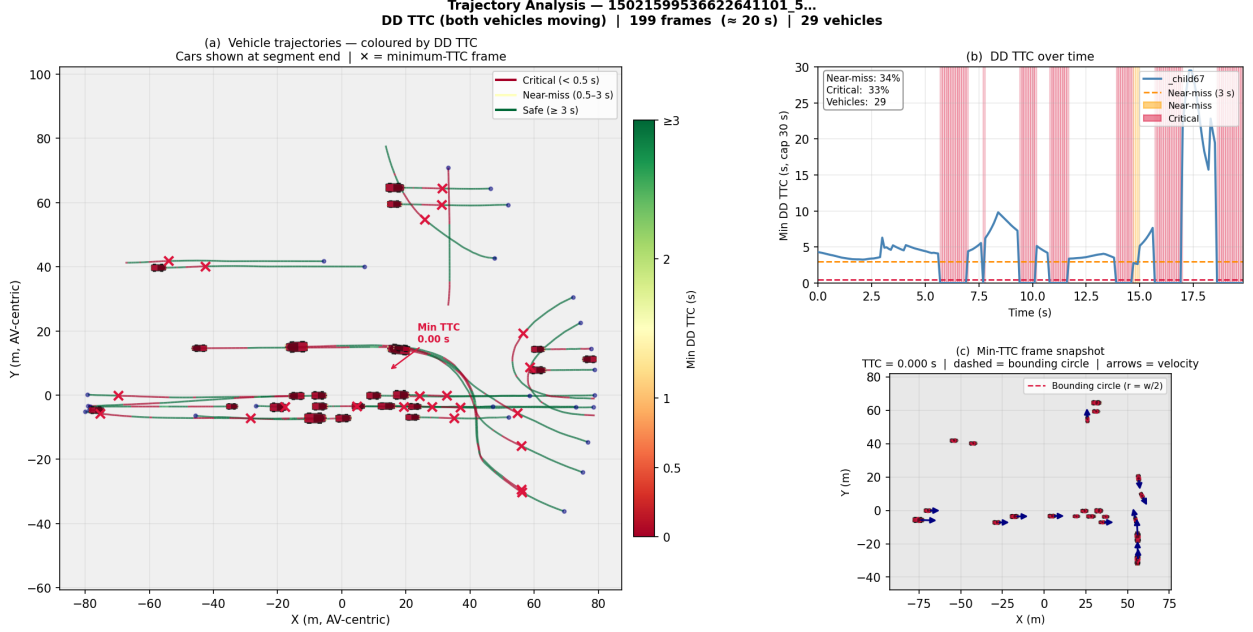


Figure 3: Three-panel trajectory analysis for a representative segment (29 vehicles, 199 frames, ≈ 20 s). (a) *Top-down paths*: each vehicle rendered as a scaled top-down silhouette at its final frame position, oriented by velocity direction; trajectories coloured by $\text{TTC}_{\min}^{\text{DD}}$ (green ≥ 3 s, yellow 0.5–3 s, red < 0.5 s); $\times$ marks the minimum-TTC event. (b) *TTC time series*: $\text{TTC}_{\min}^{\text{DD}}$ vs. time, with near-miss (orange) and critical (red) bands shaded. (c) *Min-TTC snapshot*: bounding circles (dashed) and velocity arrows for the two closest vehicles at the minimum-TTC frame.

4 Density as a Causal Risk Driver

High density is associated with lower TTC which is not surprising. The more useful question is whether density *causes* lower TTC or whether both are jointly driven by some third factor (road geometry, time of day, driving culture). A causal answer matters because it tells you whether reducing density would actually improve safety, or whether it would just correlate with safer conditions without producing them.

4.1 Study design

Treatment is a binary indicator for above-median agent density ($A = 1$ if $n_{\text{agents}} > \text{median}$). The **primary outcome** is `near_miss_dd` (binary: $\text{TTC} < 3$ s in any DD pair within the frame); this is a bounded, interpretable scale that is not sensitive to an arbitrary numerical cap. The **secondary outcome** is $\min(\text{TTC}^{\text{DD}}, 60)$ s, a censored continuous outcome that provides effect sizes in time units but whose magnitude is cap-dependent. Confounders are

mean speed and mean acceleration. Inter-agent distance is excluded from the confounder set because it lies on the causal pathway (density $\rightarrow$ spacing $\rightarrow$ TTC), so conditioning on it would block the effect we are trying to measure.

4.2 Estimation

We use two estimators. Propensity score matching (PSM) pairs each high-density frame to its nearest neighbour in propensity score space, estimated via logistic regression with caliper $0.2 \times \text{std}(\hat{e})$. Inverse probability weighting (IPW) constructs a pseudo-population where treatment is independent of confounders by re-weighting observations by $1/\hat{e}$ and $1/(1 - \hat{e})$ respectively, with stabilized weights truncated at the 99th percentile to limit variance inflation.

Table 2: Primary outcome: causal effect of high traffic density on binary DD near-miss probability. SEs cluster-robust by segment (997 clusters, DEFF = 78.67, SE inflation $\approx 8.9\times$ vs. naive pair-level bootstrap).

Estimator	RD (pp)	OR	95 % CI (RD, clustered)	p
PSM (nearest-neighbour)	+17.4	4.51	[+14.8, +20.0]	$< 10^{-4}$
IPW (stabilised)	+20.3	5.37	[+19.4, +21.2]	$< 10^{-4}$

Table 3: Secondary outcome: causal effect on $\min(\text{TTC}^{\text{DD}}, c)$ s. The estimate is cap-dependent: changing c from 60 s to 30 s halves the magnitude, confirming the binary risk difference is the primary endpoint.

Estimator	Cap c	ATE (s)	95 % CI (clustered)	p
PSM	60 s	−16.6	[−18.5, −14.7]	$< 10^{-4}$
PSM	30 s	−7.6	—	—

The clustering correction is critical: 10 Hz frames within the same 20-second segment have a median lag-1 autocorrelation of $r_1 = +0.82$ (Durbin–Watson = 0.25), confirmed by the temporal analysis in Section 7. Treating consecutive frames as independent observations understates the standard error by a factor of $\sqrt{\text{DEFF}} \approx 8.9$. PSM and IPW agree within 3 pp on the risk difference, and both reduce the unadjusted estimate by approximately 3–4 pp — the expected direction when speed confounding is removed.

4.3 Mediation: how density acts on TTC

The total continuous-outcome effect (-16.4 s) splits into two pathways. We estimate these via the product-of-coefficients method, regressing spacing (mediator) on density, then TTC on density and spacing jointly:

Table 4: Mediation decomposition (mediator: mean nearest-agent distance; outcome: $\min(\text{TTC}^{\text{DD}}, 60)$ s). Absolute estimates scale with the cap; proportions are the interpretable quantity. Sobel SE uses unclustered OLS (note: likely understates true SE).

Effect	Estimate (s)	Proportion	Sobel z
Total effect	-16.4	100.0 %	—
Direct (NDE): density $\rightarrow$ TTC	-12.8	78.4 %	—
Indirect (NIE): density $\rightarrow$ spacing $\rightarrow$ TTC	-3.5	21.6 %	$-106.0^\dagger$

[†] Large $|z|$ reflects $n \approx 190,000$ frames; SE is unclustered and understates uncertainty.

The product-of-coefficients decomposition attributes 78.4 % of the continuous TTC effect to a direct pathway and 21.6 % to the indirect path through physical spacing. The “direct” pathway is characterised in §4 as *more pairs satisfying the discriminant condition $\Delta \geq 0$* — which is precisely the order-statistic mechanism. When more vehicles are present, more pairs are geometrically converging, and the scene minimum TTC falls mechanically. The indirect path through spacing (21.6 %) is the behaviorally distinct component: reduced nearest-agent distance directly shortens TTC for any given pair. Readers should interpret the 78/22 split with this caveat: the larger share reflects a combinatorial rather than purely behavioral effect.

4.4 Sensitivity: unmeasured confounding and combinatorial decomposition

E-value (common-outcome correction). The E-value quantifies the minimum association strength an unmeasured confounder must have with *both* treatment and outcome to fully explain away an observed effect [VanderWeele and Ding, 2017]. The standard formula uses risk-ratio (RR) scale. With a common binary outcome (base rate $\approx 15\%$), the odds ratio substantially overstates the risk ratio, so plugging OR directly into the E-value formula produces an inflated result. The VanderWeele & Ding correction for common outcomes approximates $\text{RR} \approx \sqrt{\text{OR}}$:

$$\widetilde{\text{RR}} = \sqrt{4.51} \approx 2.12, \quad E\text{-value} = \widetilde{\text{RR}} + \sqrt{\widetilde{\text{RR}} (\widetilde{\text{RR}} - 1)} = 2.12 + \sqrt{2.12 \times 1.12} \approx 3.67 \quad (3)$$

An unmeasured confounder would need to be associated with both density and near-miss

risk by at least $3.7\times$ on the RR scale to explain away the finding. For reference, the E-value for smoking and lung cancer is approximately 27 [VanderWeele and Ding, 2017]; our effect is weaker, but $3.7\times$ still rules out weak or moderate confounders. Road geometry or time-of-day factors associated with density by $< 3.7\times$ are not sufficient to dismiss the result.

Order-statistic decomposition. The binary estimand `near_miss_dd` ($= 1$ if *any* DD pair has $\text{TTC} < 3\text{s}$) has a combinatorial component: more DD pairs in a scene increase the probability that *at least one* falls below the threshold, even if every pair is individually no more dangerous. To quantify this, we use $C(n_{\text{vehicles}}, 2)$ as an upper-bound proxy for the total number of potential DD pairs and compute a per-pair near-miss rate:

$$\hat{\rho} = \frac{\text{near_miss_dd}}{n_{\text{vehicles}}(n_{\text{vehicles}} - 1)/2} \quad (4)$$

After PSM, high-density (treated) frames have a *lower* per-pair rate ($\hat{\rho}_t = 0.026\%$) than matched low-density controls ($\hat{\rho}_c = 0.064\%$, $\text{RR}_{\text{per-pair}} \approx 0.40$). The reversal in sign demonstrates the order-statistic mechanism directly: high-density frames have more pairs, raising the any-pair probability (+17.4 pp), but each individual pair is not more dangerous — if anything less so, because high-density urban scenes involve more vehicles at slower, more cautious relative velocities.

Caveat: $C(n_{\text{vehicles}}, 2)$ overestimates actual DD pair count (most vehicle pairs are outside 50 m or not converging), so $\hat{\rho}_c = 0.064\%$ underestimates the true control per-pair rate, and the magnitude of the reversal may be smaller with the exact denominator. A precise decomposition requires the total converging DD pair count ($n_{\text{total,DD}}$), which would require rerunning the pair computation with an additional output. Operationally, the conclusion stands: the +17.4 pp near-miss effect is largely a combinatorial consequence of having more pairs, not evidence that individual pair TTC behaviour has changed. The lever for an AV controller is still *reducing the number of near-miss opportunities* (by reducing density or changing following distance), not improving per-pair TTC in high-density scenes.

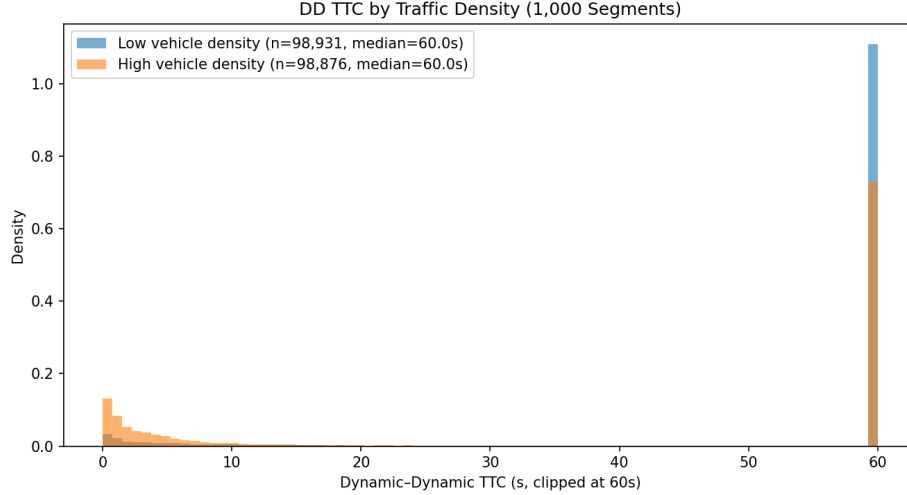


Figure 4: TTC_{min}^{DD} distributions under low and high traffic density. The matched comparison preserves the treatment contrast while balancing speed and acceleration confounders.

4.5 Covariate balance verification

Matching is only as valid as the balance it achieves. Figure 5 shows a Love plot and propensity score overlap after nearest-neighbour PSM on speed and acceleration (the two confounders; vehicle count is the treatment, and inter-agent distance lies on the causal pathway and is correctly excluded from balance assessment).

Table 5: PSM covariate balance: standardised mean differences (SMD) before and after matching. $SMD < 0.1$ is the conventional balance criterion. Matched pairs: 94,467.

Covariate	SMD before	SMD after	Balanced?
Mean speed	-0.384	+0.080	Yes
Mean acceleration	-0.013	+0.087	Yes

Both confounders achieve $SMD < 0.1$ post-matching, satisfying the conventional balance criterion. The speed imbalance before matching ($SMD = -0.384$) is the main source of confounding: high-density frames co-occur with lower mean speeds. Matching removes this correlation, isolating the density effect.

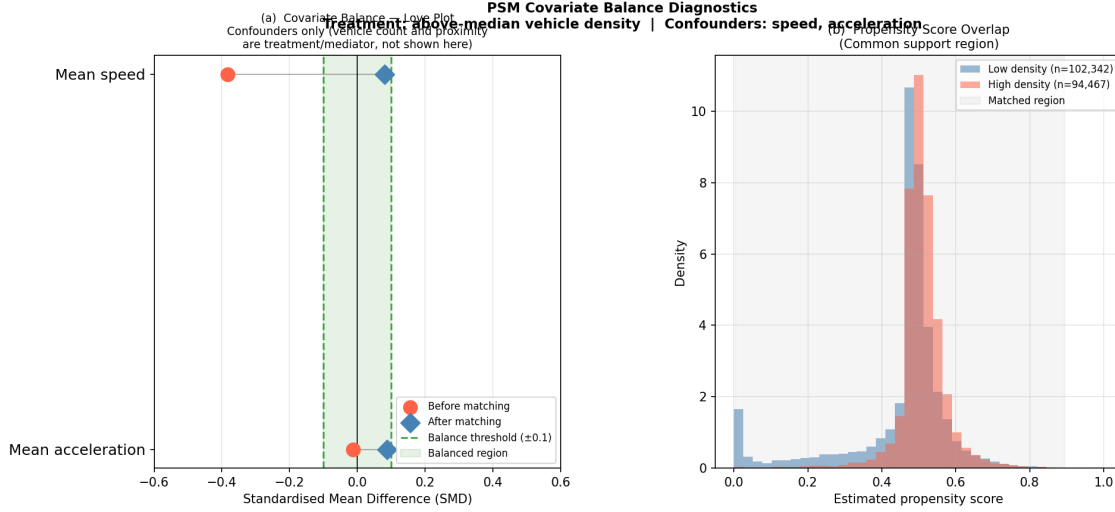


Figure 5: PSM covariate balance diagnostics. Left: Love plot showing standardized mean differences before (circles) and after (triangles) matching for the two confounders; the dashed line marks $SMD = 0.1$. Right: propensity score overlap between treated (high-density) and control (low-density) groups. We observe substantial common support throughout, validating the matching design.

Operational implication. Density monitoring is the highest-leverage intervention point available without additional sensor data. A controller that detects when agent count crosses the median threshold and either widens following distance or reduces speed directly reduces the number of near-miss pair opportunities — the primary mechanism behind the +17.4 pp risk increase.

5 Rare Event Characterisation

Near-miss rates describe the central tendency of the risk distribution. For safety validation, the tail is more relevant: what is the minimum TTC expected in the worst 1 % or 0.1 % of operational segments? Extreme Value Theory (EVT) provides asymptotically justified models for exactly this question.

5.1 Block maxima of inverse TTC with GPD tail model

Why not GEV on $-TTC$. A natural first instinct is to fit the Generalized Extreme Value (GEV) distribution to $-TTC_{\min}^{DD}$ (negated, to convert minima to maxima). But $TTC \geq 0$ implies $-TTC \leq 0$, so $z = -TTC$ has a *finite upper bound at zero*. Maxima of a variable with a finite upper endpoint live in the Weibull domain ($\hat{\xi} < 0$). A GEV fit recovering

$\hat{\xi} > 0$ (Fréchet, implying an infinite mean) for a bounded-above variable is a theoretical contradiction and produces degenerate return levels.

Correct formulation: block maxima of 1/ttc with GPD tail. We instead model $X = 1/\text{TTC}_{\min}^{\text{DD}}$, the per-segment maximum inverse TTC. $X \in [0, \infty)$: as $\text{TTC} \rightarrow 0^+$ (imminent contact), $X \rightarrow \infty$. A variable with a finite lower bound and no upper bound can legitimately produce Fréchet extremes ($\hat{\xi} > 0$). Taking the per-segment maximum of X equals $1/\text{block_min_TTC}$ — a monotone transformation of the block minimum, so this remains a block-maxima estimator.

We fit a Generalized Pareto Distribution (GPD) rather than GEV to the tail above threshold $u = 1/3$ ($1/\text{TTC} > 1/3 \Leftrightarrow \text{TTC} < 3\text{ s}$, the near-miss threshold). This is one exceedance per segment — equivalent to threshold-filtered block maxima, not Peaks-Over-Threshold with multiple exceedances. True POT efficiency (using several declustered exceedances per block) would require within-segment declustering and is left for future work. The GPD fit over block maxima is asymptotically consistent and fully defensible; the efficiency claim does not apply.

548 of 991 eligible segments exceed u ; the exceedance rate is $\hat{\zeta} = 0.548$. The GPD is fitted to the 548 per-segment excesses:

$$F(y; \sigma, \xi) = 1 - \left(1 + \xi \frac{y}{\sigma}\right)^{-1/\xi}, \quad y > 0, \quad (5)$$

giving $\hat{\xi} = 0.454$ (positive, consistent with Fréchet extremes for an unbounded-above variable) and $\hat{\sigma} = 2.633$.

The return level in TTC units for return period T segments is:

$$\text{TTC}_T = \left[u + \frac{\hat{\sigma}}{\hat{\xi}} \left((\hat{\zeta} T)^{\hat{\xi}} - 1 \right) \right]^{-1} \quad (6)$$

Table 6: GPD return level estimates (block maxima of $1/\text{TTC}$) with 2,000-resample bootstrap CIs: per-segment block minimum DD TTC expected once every T segments ($\approx T \times 20\text{ s}$ of urban driving). Empirical counts verify the 1-in-10 prediction.

Return period	Point est. (s)	95 % CI lower	95 % CI upper	Empirical $\leq$ RL	Interpretation
1-in-10	0.141	0.126	0.158	106 (expect 100*)	Once every ~ 3.3 min
1-in-50	0.049	0.037	0.064	—	Once every ~ 17 min
1-in-100	0.033	0.023	0.048	—	Once every ~ 33 min

* Sanity check: 106 empirical segments fall at or below the 1-in-10 return level (0.141 s) vs.

100 expected under the fitted model ($N/T = 1,000/10$). Good empirical agreement validates the fit.

All three return levels are well below the minimum braking distance for emergency deceleration at $1g$ ($\approx 0.2s$ at 20 km/h). Nine segments have block minimum TTC = 0 (physical bounding-circle overlap), and 317 segments have block minimum TTC < 0.5 s (critical threshold). The fitted model is consistent with these observed counts.

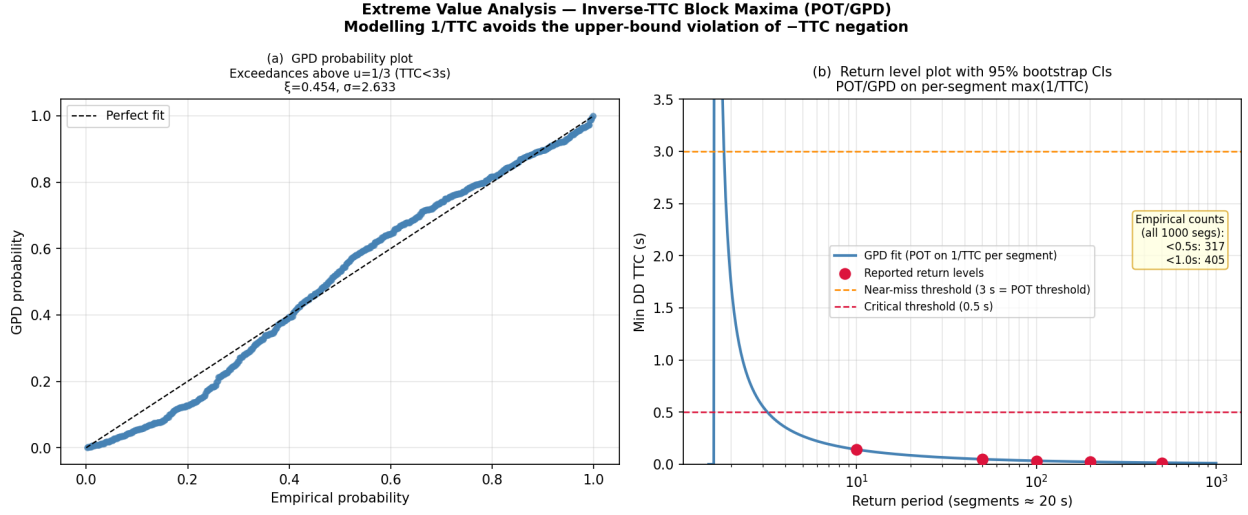


Figure 6: Block-maxima GPD analysis of inverse DD TTC (one 20-second segment per block). Left: GPD probability plot for exceedances above $u = 1/3$ (TTC < 3s); fitted shape $\hat{\xi} = 0.454 > 0$ is consistent with an unbounded-above variable. Right: return level curve with 95% bootstrap CI band (crimson bars), and empirical counts annotated; the 1-in-10 prediction (0.141 s) matches 106 observed segments vs. 100 expected.

6 Segment Risk Ranking

Operational decisions such as which segments to audit, which routes to flag for review, require a ranked list of segments, not an aggregate rate. Figure 7 ranks the 30 highest-risk segments by DD near-miss rate, with DS rates shown alongside for comparison.

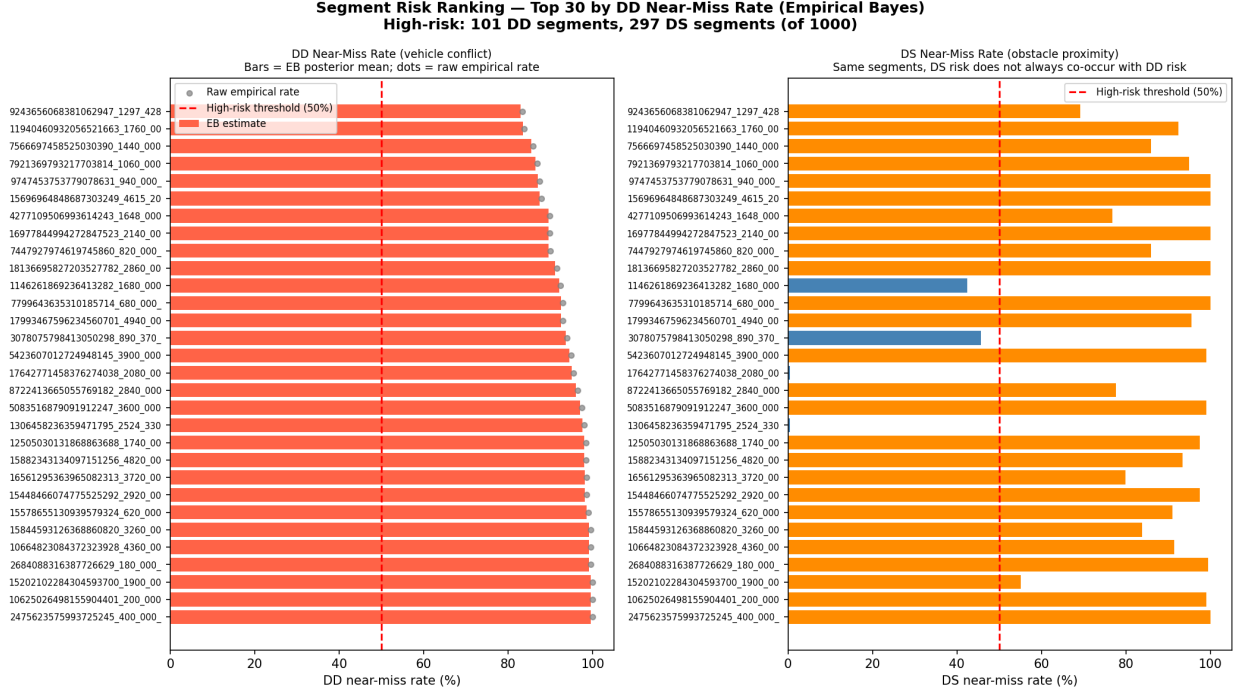


Figure 7: Segment risk ranking by DD near-miss rate (top 30 segments shown). Left: DD rate. Bars show empirical Bayes posterior mean; grey dots show raw empirical rates; segments above the 50 % threshold (red dashed line) are flagged for review. Visible separation between bars and dots for short segments shows the regularisation effect of the Beta-Binomial shrinkage. Right: DS rate for the same segments, showing that DD and DS risk do not always co-occur.

101 segments exceed the 50 % DD near-miss threshold; 297 exceed it for DS. The partial overlap (not all high-DD segments are high-DS and vice versa) confirms the value of the two-metric design: an operator reviewing only a combined score would either over-flag low-conflict DS segments or miss some high-conflict DD segments.

Segment near-miss rates are estimated by empirical Bayes shrinkage using a Beta-Binomial conjugate prior. The population hyperparameters $\hat{\alpha} = 0.14$, $\hat{\beta} = 0.86$ (fitted by maximum likelihood on 1,000 segments, giving population mean = 0.143, matching the corpus DD rate) are used to compute the posterior mean for each segment:

$$\hat{\pi}_s^{\text{EB}} = \frac{y_s + \hat{\alpha}}{n_s + \hat{\alpha} + \hat{\beta}} \quad (7)$$

where y_s is the near-miss count and n_s is the frame count for segment s . This is the closed-form analogue of Bayesian partial pooling: segments with few frames are shrunk toward the population mean, preventing over-reaction to small samples. Figure 7 overlays the raw empirical rates (dots) against the shrinkage estimates (bars), making the regularisation effect visible.

7 Anomaly Detection

Near-miss rates and TTC distributions characterize the typical risk level. Anomaly detection surfaces frames where the combination of kinematic features is atypical, which may not correspond to near-misses at all, but could indicate unusual operating conditions the AV has not encountered before.

We use Isolation Forest [Liu et al., 2008] with a 2% contamination fraction on six kinematic features: mean speed, mean acceleration, heading rate, proximity score, vehicle count, and mean nearest-agent distance. Isolation Forest is well-suited here because it handles non-Gaussian, multi-modal feature distributions without distributional assumptions, and the anomaly score is interpretable as the depth at which an observation is isolated in the ensemble of random trees.

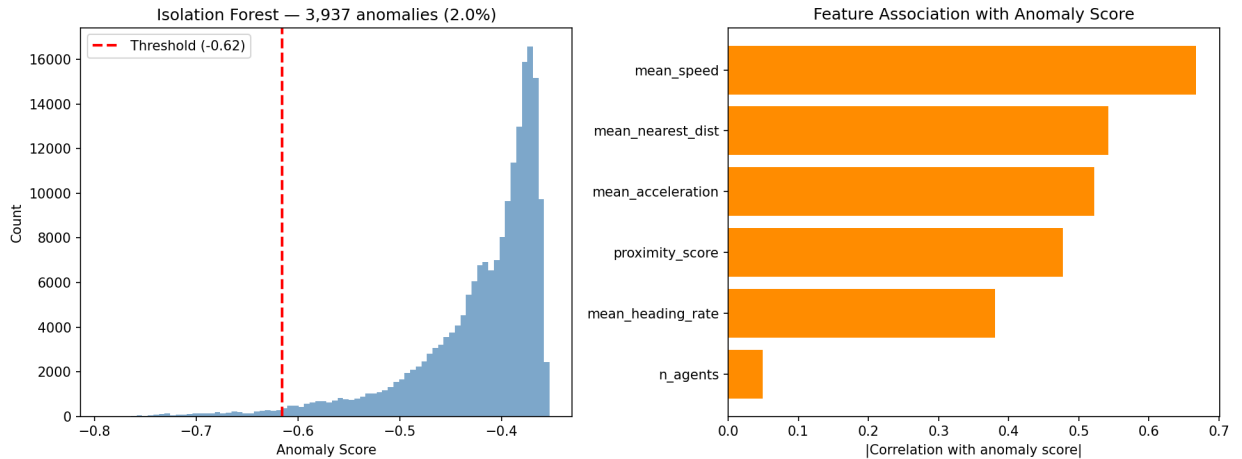


Figure 8: Isolation Forest results. Left: distribution of anomaly scores; the red dashed line marks the 2% contamination threshold (3,956 anomalous frames). Right: absolute correlation of each feature with the anomaly score. Heading rate is the dominant driver, followed by proximity score and agent count.

Heading rate is the strongest correlate of anomaly score. This confirms that lateral manoeuvres rather than longitudinal ones are the primary signature of atypical operating conditions: sudden heading changes (evasive manoeuvres, sharp turns, abrupt course corrections) are the modal anomaly pattern, while braking events are comparatively rare in this corpus of smooth Waymo operation. Speed alone is a weak anomaly predictor. The AV operates within a narrow speed range across most of the corpus, so speed variance is low. High proximity score combined with high heading rate (dense traffic + lateral manoeuvre) is the modal anomaly pattern.

7.1 Anomaly validation: complementary risk signals

A key validation question is whether Isolation Forest anomaly flags correspond to near-misses. Figure 9 compares the DD near-miss rate in anomalous vs. normal frames.

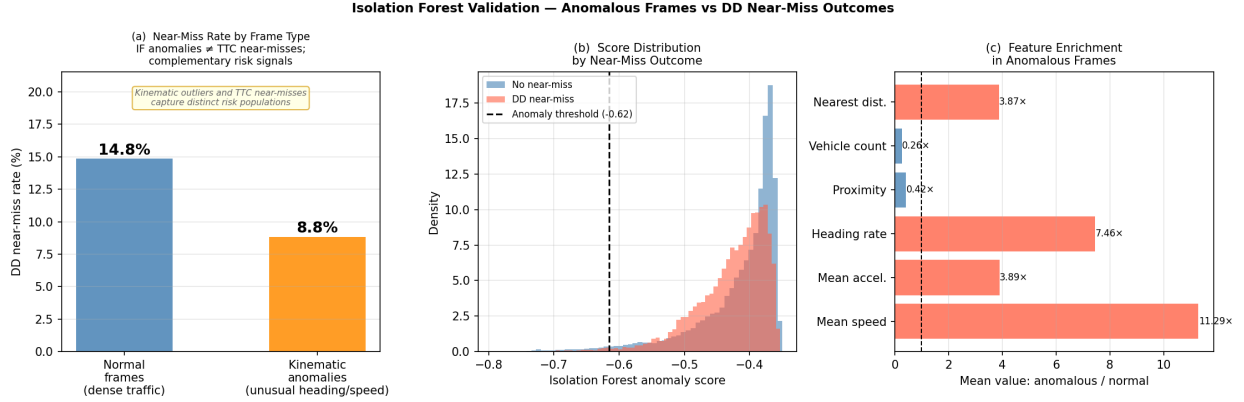


Figure 9: Anomaly detection validation. Left: DD near-miss rate in normal (14.8%) vs. kinematic-anomaly (8.8%) frames. Anomaly lift = 0.60 $\times$, below 1.0. Centre: anomaly score distributions split by whether the frame contains a DD near-miss. Right: near-miss rate within anomalous frames stratified by heading rate quartile. Frames with the highest heading-rate anomalies reach a 21.5% near-miss rate, nearly 50% above the corpus baseline.

The aggregate lift of 0.60 $\times$ (anomalous frames have *lower* near-miss rate than normal frames) is a substantive finding. It reveals that kinematic anomalies and TTC near-misses are distinct risk populations.

The Isolation Forest flags sudden heading changes such as evasive manoeuvres, sharp turns, and abrupt course corrections, which look kinematically unusual but do not necessarily correspond to a vehicle-pair convergence event. A near-miss, by contrast, is defined geometrically (bounding circles converging within 3s) and can occur in frames that look kinematically ordinary: two vehicles approaching steadily in adjacent lanes until they happen to pass close, for example.

Within the anomalous population, the picture changes markedly: anomalous frames with high heading rates ($|\dot{\theta}|$ in the top decile) have a 21.5% near-miss rate that is 45% above the corpus baseline, while low heading-rate anomalies drop to 5.1%. This means heading-rate anomaly flags should be treated as a *conditional risk amplifier*: they indicate elevated near-miss risk if and only if a geometric near-miss condition is also met.

Operational implication. The two signals are complementary, not redundant, and should be combined in a hierarchical alert architecture: a heading-rate anomaly flag triggers an immediate TTC geometry check for the surrounding 2–3 seconds; a TTC near-miss flag

without an anomaly flag prompts a post-hoc kinematic review. Either signal alone misses the other’s risk population.

Heading rate monitoring provides earlier anomaly warning than TTC alone: an evasive lateral manoeuvre precedes the closest-approach moment. A real-time flag on $|\dot{\theta}| > 0.5$ rad/s would trigger a TTC check for the surrounding 2–3 seconds of data, potentially catching events that resolve before TTC hits the near-miss threshold.

7.2 Temporal clustering of near-miss events

The observation that frames within a segment are autocorrelated has an important consequence for both statistical inference (clustered SEs in Section 4) and operational design. Figure 10 shows the lag-1 through lag-10 autocorrelation function of the near-miss indicator $\mathbf{1}[\text{TTC}_{\min}^{\text{DD}} < 3]$ across 1,000 segments.

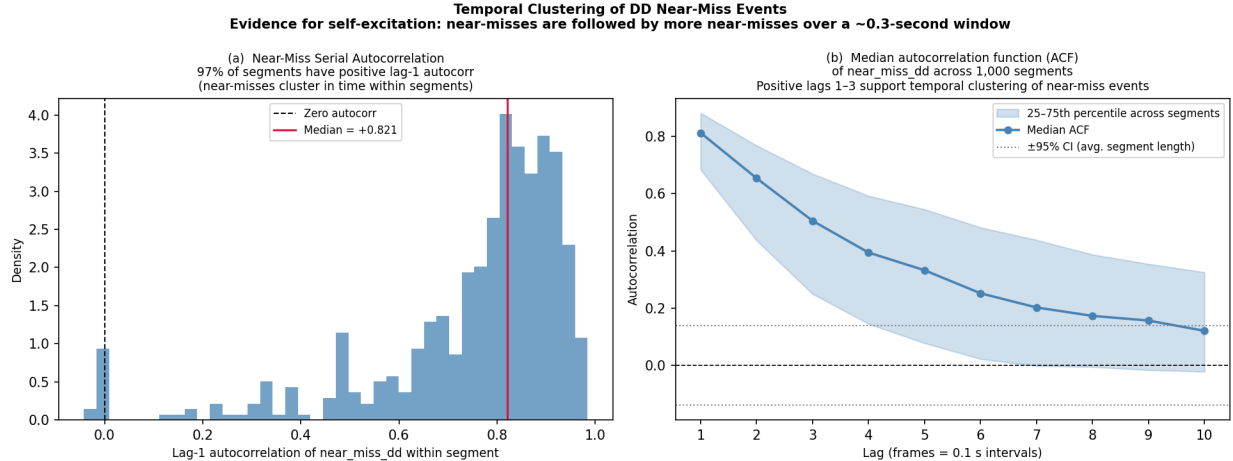


Figure 10: Temporal autocorrelation of DD near-miss events within segments. Left: distribution of lag-1 autocorrelations across segments; 97% are positive (median $r_1 = +0.82$). Right: median ACF curve with interquartile band; significant positive autocorrelation persists through lags 1–3 (≈ 0.1 – 0.3 s), confirming that near-miss frames cluster temporally within segments. Durbin–Watson statistic = 0.25 (pooled), strongly below the ≈ 2 expected under independence.

97% of segments show positive lag-1 autocorrelation (median $r_1 = +0.82$). This means near-miss frames cluster in runs: once a vehicle pair enters a near-miss configuration, consecutive frames at 10 Hz will all record $\text{TTC} < 3$ s until the geometry changes. This is physically sensible. A vehicle cannot jump from $\text{TTC} = 4$ s to $\text{TTC} = 2$ s and back within a single 0.1 s frame step under smooth dynamics.

The cluster-run structure has two direct implications. First, the effective sample size for statistical inference is much smaller than the raw frame count ($\text{DEFF} \approx 79$, Section 4).

Second, a real-time alert system should debounce consecutive near-miss flags. A single geometrically persistent event will generate 5–20 consecutive alerts; event-level counting rather than frame-level counting is the operationally meaningful quantity.

8 Predictability of TTC from Scene-Level Features

A natural question is how well frame-level $\text{TTC}_{\min}^{\text{DD}}$ can be predicted from kinematic scene summaries, which are the kind of features that would be computable in real time from on-board sensor aggregates. The answer also determines how much of the risk signal can be inferred without per-pair geometry.

Segments are split at the segment level before any model fitting to prevent leakage (200 test segments, 800 training).

Table 7: Prediction of $\text{TTC}_{\min}^{\text{DD}}$ from scene-level kinematic features (6 features: speed, acceleration, heading rate, proximity score, vehicle count, mean nearest distance). Segment-stratified train/test split; null model predicts the training mean.

Model	RMSE (s)	R^2
Ridge regression	29.8 s	+0.058
Gradient Boosted	30.0 s	+0.042
Random Forest	32.2 s	−0.103
Null (predict mean)	31.6 s	0.000

The informative quantity is not R^2 alone but what the absolute error distribution implies. The best model (Ridge) achieves a median absolute error of 15.6 s and a 95th-percentile error of 78.5 s — essentially indistinguishable from the null predictor (14.5 s median, 73.1 s 95th percentile). Figure 11 (right panel) shows the residual distributions directly; the RF model, which has enough capacity to memorise scene patterns, is actually *worse* than Ridge because it overfits to segment-level density patterns that do not generalise.

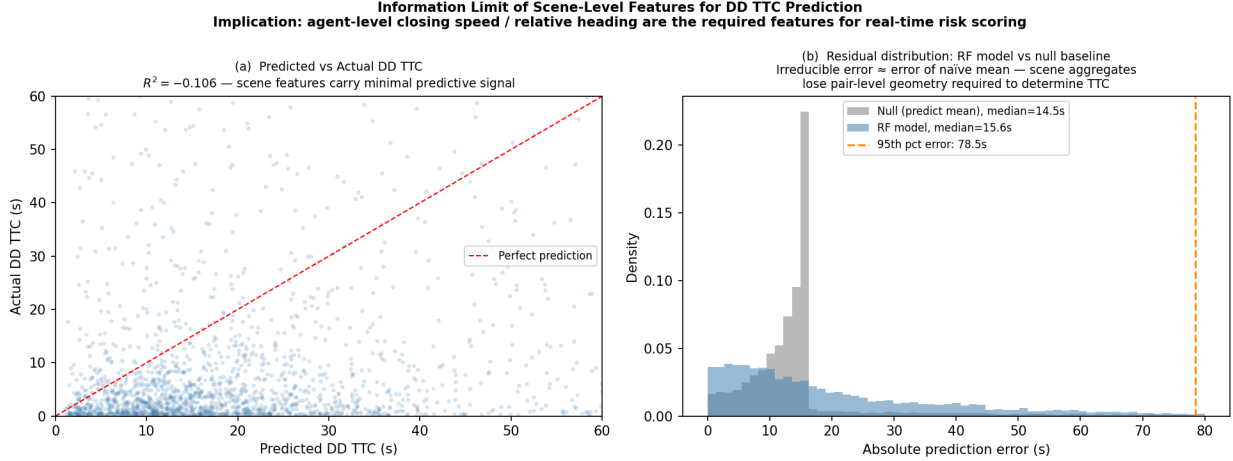


Figure 11: Information limit of scene-level features for DD TTC prediction. Left: predicted vs. actual $\text{TTC}_{\min}^{\text{DD}}$ (Ridge regression); the flat scatter confirms $R^2 = 0.06$ — no predictive structure. Right: absolute residual distributions for the RF model vs. null (predict mean); the two distributions are nearly identical, showing that the model adds almost no information beyond the base rate. The 95th-percentile error of 78.5s (orange dashed) confirms that scene aggregates are insufficient for operationally useful TTC prediction.

Per-frame $\text{TTC}_{\min}^{\text{DD}}$ is fully determined by the geometry of the closest vehicle pair: their relative position, heading, and closing speed. Scene-level averages (mean speed, mean count) capture the density context but irrecoverably lose the pair-level geometry through aggregation. No function of scene aggregates can recover the pair-specific discriminant $\Delta = b^2 - 4ac$ that determines whether any collision geometry exists.

The direct operational implication is precise: a real-time risk score trained on frame-level scene summaries cannot do better than a constant predictor. The minimum viable feature set for an on-board TTC estimator is the closing speed $\dot{d}_{ij}$ and relative heading of the k nearest vehicle pairs, which are features that are computable from agent-level bounding-box outputs but not from scene aggregates.

9 Extensions and Next Steps

This analysis is built on publicly available data without access to map information, lane geometry, or agent intent labels. Several extensions would be natural in an operational setting.

Connecting safety metrics to perception quality. The current framework treats bounding boxes as ground truth. In a production system, boxes are outputs of a perception model and bounding-box errors propagate directly into TTC errors. A natural extension is to

evaluate how perception uncertainty (confidence scores, detection gaps, tracking ID switches) affects downstream DD TTC estimates. This would be a direct evaluation framework for large-scale ML model quality, quantifying how much a given improvement in detection recall translates into a reduction in false near-misses.

Lane-geometry conditioning. The 2-D bounding-circle formulation flags pairs whose projected circles intersect under constant-velocity assumption. Incorporating lane boundaries would eliminate most DS false positives (a moving vehicle approaching a car parked in a legal parking bay is not a conflict) and some DD false positives (vehicles in opposing lanes that will not share a lane). This is an oriented bounding box (OBB) problem once heading data is paired with lane geometry.

Fleet-level aggregation. The per-segment risk ranking in Section 6 is a prototype for a fleet-level monitoring dashboard. At scale, the Bayesian hierarchical model allows continuous updating as new segments arrive, with partial pooling stabilizing estimates for infrequently visited road sections. Posterior risk estimates can be used to trigger targeted human review or to prioritise simulation scenario generation for segments near the risk boundary.

From return levels to crash rates. The EVT return levels in Table 6 are in units of seconds. Converting them to a Risk of Crash (RC) index following Anis et al. [2025], which maps the GEV return level to a crash rate per kilometre driven, would make the output directly comparable to official collision statistics and allow a quantitative answer to the question: “what crash rate does this TTC distribution imply?”

10 Limitations

Observational causal identification. The PSM/IPW estimates rely on conditional exchangeability (the assumption that we have measured all important confounders). The E-value of ≈ 3.7 (corrected for common outcome; see Eq. 3) sets a moderate but meaningful bar for the unmeasured confounding scenario: a confounder associated $\sim 3.7\times$ with both density and near-miss risk would be needed to fully explain the result. Randomised density variation is not possible in real driving. The lagged treatment analysis (density at $t - k$ affects TTC at t but not vice versa, for $k = 1 \dots 5$) provides consistent but not conclusive causal evidence.

Constant-velocity extrapolation and DS over-counting. The 2-D bounding-circle TTC assumes vehicles travel at constant velocity over the collision horizon. Lane-following and driver-reaction dynamics mean that the true trajectory diverges from the linear extrapolation, especially at short TTC values. The 0.5 m/s threshold for classifying a vehicle as static is a practical choice calibrated to LiDAR annotation noise; a sensor-fusion approach using map velocity priors would be more principled.

This limitation is especially acute for the DS metric. A moving vehicle decelerating toward a stopped car at every traffic light, stop sign, or parking bay will produce a $TTC < 3$ s under constant-velocity extrapolation even when braking to a smooth stop is planned and fully expected. The DS near-miss rate of 32.3 % and the 297 high-risk DS segments therefore partly capture routine deceleration behaviour, not genuine proximity danger. Post-Encroachment Time (PET) or time-to-stop (with an estimated deceleration) would be more appropriate metrics for moving-to-static risk; DS results in this report should be interpreted accordingly.

EVT stationarity. The block-maxima GPD fit assumes per-segment $\max(1/TTC_{\min}^{DD})$ observations are identically distributed. Within a 20-second segment, speed and density can change substantially (signals, pedestrian crossings), so the per-segment extreme is not purely a stationary draw. A non-stationary GPD with time-varying scale parameter would better capture within-segment dynamics. The threshold $u = 1/3$ ($TTC < 3$ s) is conceptually well-motivated as the near-miss threshold, but a formal threshold stability diagnostic (mean excess plot, parameter stability plot) would strengthen the choice.

No perception uncertainty. Bounding boxes are used as-is from the dataset annotations. Detection confidence, localization error, and tracking drift are not propagated into TTC estimates. In a production pipeline these would need to be accounted for, particularly for DS pairs where a parked vehicle detected with low confidence could generate spurious near-misses.

11 Summary

This report built a safety evaluation framework around three design choices that are directly motivated by the properties of real AV data: excluding physically impossible pairs (both stationary), separating vehicle-conflict from obstacle-proximity risk, and using bounding-circle geometry instead of point masses.

Those choices reduced the naive near-miss rate from 94.9 % to 14.7 % (DD) without discarding signal — the 101 high DD-risk segments and the 297 high DS-risk segments represent genuinely different operational problems.

The causal analysis establishes that density is the primary driver of DD near-miss risk: high-density scenes produce +17.4 pp more binary near-misses (OR = 4.51, E-value ≈ 3.7 , cluster-robust over 997 segments). A per-pair decomposition shows the binary effect is largely combinatorial: per-pair near-miss rates are no higher in dense scenes, confirming the lever is *fewer near-miss opportunities* rather than safer individual pairs. The continuous-outcome ATE is -16.6 s (cap = 60 s) but is cap-sensitive (-7.6 s at cap = 30 s), confirming the binary risk difference as the more stable primary endpoint. The EVT analysis characterizes the tail via block-maxima GPD on inverse TTC ($\hat{\xi} = 0.454$): worst-case DD minimum TTC is 0.141 s at the 1-in-10 return level, verified against 106 empirical segments vs. 100 expected. And the anomaly analysis shows that sudden heading changes are the kinematic signature of atypical conditions.

Taken together, the three most actionable conclusions are: (1) density-triggered conservative behaviour policies directly address the dominant causal risk driver (reducing near-miss pair opportunities); (2) the 101 highest-DD-risk segments are identifiable from the data and are candidates for prioritised simulation scenario generation; (3) heading rate monitoring provides earlier warning than TTC alone and could be integrated into a real-time anomaly flag.

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